Electronic Calculator Project Lab Report

ELT-210-20459
Electronic Fabrication

Instructor: Chiou, J

Team: Group 4

Report Written: 5/3/2024

School of Sciences and Engineering Department of Electrical & Computer Engineering



Group 4 - New Jersey Instruments Tech
ELT-210
5/3/2024

Lessons Learned

- Schematics were incorrect and needed to be corrected
- Parts shipment was longer than expected. Temporary parts were needed for testing/ prototyping
- Harsh knows what a tree looks like
- Certain parts needed to be modified for assembly. This included soldering pins onto the keypad and power switch
- 9 volt battery doesn't stay full charged for long and needs to read over 9.3 volts for prototype to work
- Ardino code was initially downloaded incorrectly and needed to be redownloaded
- Don’t choose impossible project

Task Number	Task	Duration	Date	Staff
10	Create Team	0.5 Hr	January-24	Team
20	Planning	1 Hr	January-24	Team
30	Research	0.5 Hr	January-24	Team
40	Design	1 Hr	January-24	Team
50	Order Parts	1 Hr	February-24	Harsh
60	Schematics	3 Hrs	February-24	Andrew
70	Soldering Work	3 Hrs	February-24	Team
80	Testing	4 Hrs	February-24	Team
90	Coding	2 Hrs	March-24	Harsh
100	3d Model Housing	1 Hr	March-24	Gabriel
110	3d Print Housing	1 Hr	March-24	Gabriel
120	Assemble	1 Hr		Team
130	Prototype Presentation	1 Hr	March-24	Team
140	PCB Practice	0.5 Hr	March-24	Josue
150	PCB Layout	2 Hrs	3/29/24	Josue
160	Budgeting	1 Hr	3/29/24	Harsh
170	Write Documentation	5 Hrs		Team
180	Test	2 Hrs	-	Team
190	Review/ Fix	1 Hr	-	Team
200	Repairs	2 Hrs	-	Team
210	Final Presentation	0.5 Hr	5/24/24	Team

Task Number	Task	Duration/ Cost	Date	Staff	Predecessor	Successor	Hand In's	Hand Out's
10	Create Team	0.7hrs	1/19/2024	Project Manager	N/A	20	RFP	Staff List
20	Pitch Ideas	1.0hrs	1/19/2024	Whole Team	10	30		
30	Select Idea	1.3hrs	1/19/2024	Project Manager	20	40		Team Name
40	Create Plan	0.5hrs	1/26/2024	Project Planner	30	50		
50	Staff Plan	0.5hrs	1/26/2024	Project Planner	40	60		
60	Present Plan	0.1hrs	1/26/2024	Project Manager	50			Project Plan
70	Research Product	4.0hrs	01/19/24-02/09/24	Lead Engineer	30	80, 100		
80	Schematic	8.0hrs	2/2/2024	All Engineers	70	90, 100, 110		
90	Layout	8.0hrs	2/2/2024	All Engineers	80	100, 110		
100	Order Parts	1.0hrs	2/2/2024-02/16/24	Procurement	70, 80, 90	110		
110	Prototype	16.0 hrs	2/9/2024	All Engineers	80, 90, 100	120		
120	Test and troubleshoot	16.0 hrs	02/09/24-02/23/24	All Engineers	110	130		
130	Present Prototype	0.1 hrs	3/1/2024	Project Manager	120			Prototype Milestone
140	Status Reports	2.0 hrs	01/19/24-03/01/24	Whole Team				



Group 4 - New Jersey Instruments Tech
ELT-210
5/3/2024

Bill Of Materials

- Keypad $5.95
- Switch $0.73
- LCD Display $6.76
- Arduino $26.30
- Potentiometer $3.82
- 9v Battery $1.49
- Battery Connector $0.52
- Labor $0.00
- 3d Printing $4.00
- PCB FREE
- Total $54.12

Building Instructions

1. Arduino

a. Loading Code

1. Acquire an Arduino Uno SMD R3.
2. Upload the NJIT Calculator code by plugging the USB cable into the Arduino and a computer.

b. Wiring

1. The Arduino works as the brain of the calculator.

2. LCD Display

a. Wiring

1. Pin 8 to 13 on Arduino.
2. Connect resistor, potentiometer.
3. Connect 5 volts and ground.
4. Follow the instructions diagram.

3. Keypad

a. Wiring

1. Connect keypad to arduino pins 0 to 7

4. Potentiometer

a. Tuning

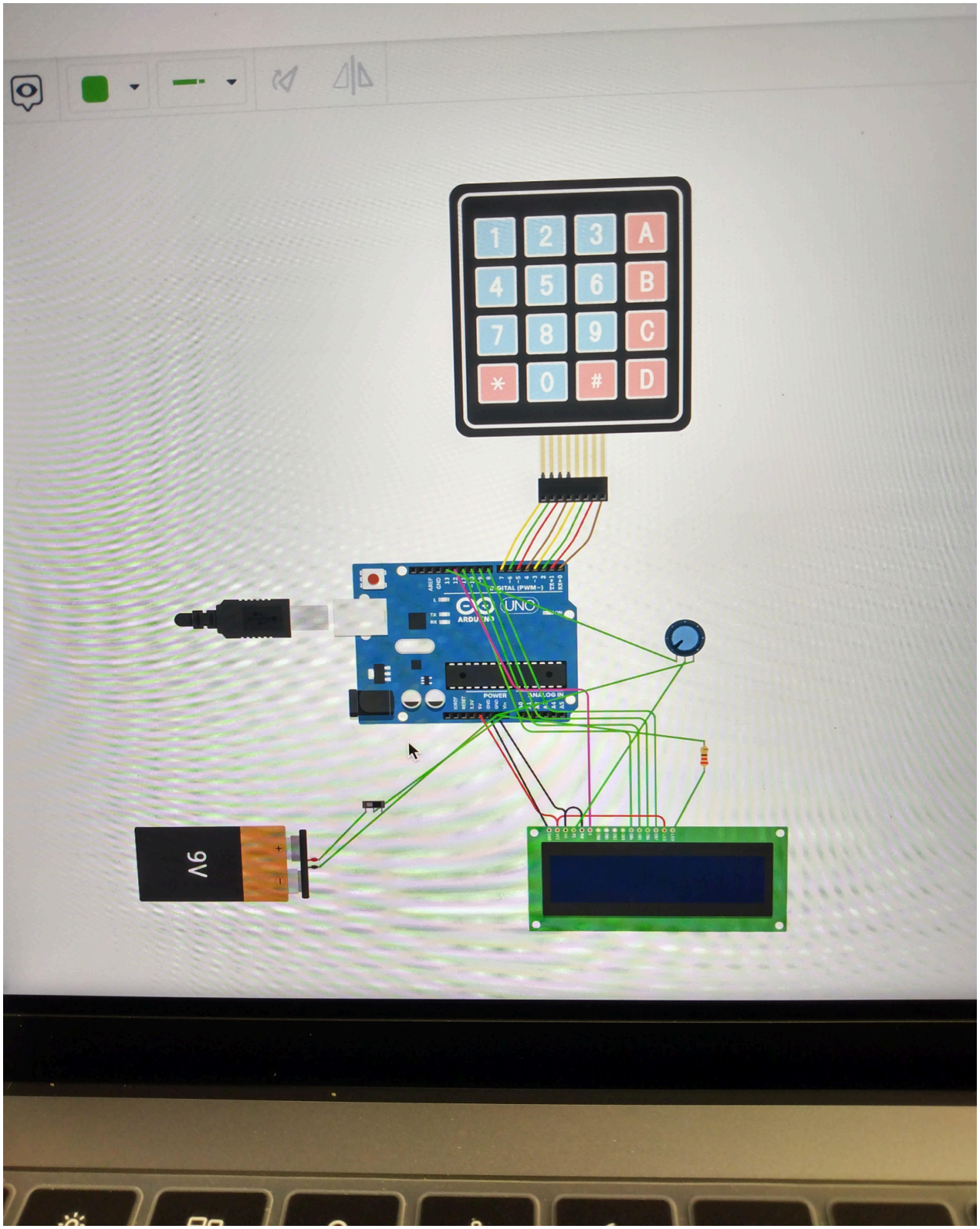
1. Adjust the potentiometer until the LCD screen display is visible.

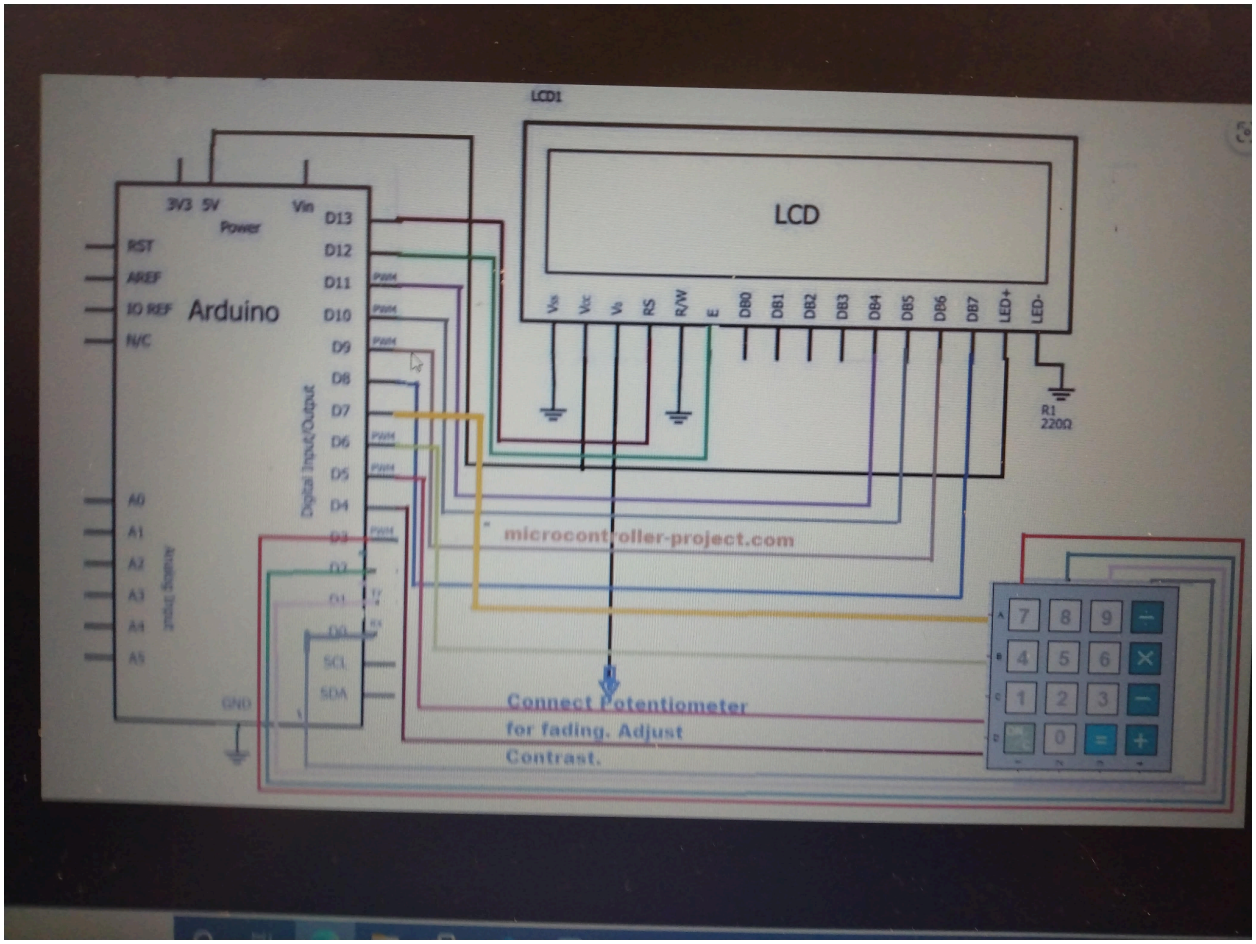
b. Wiring

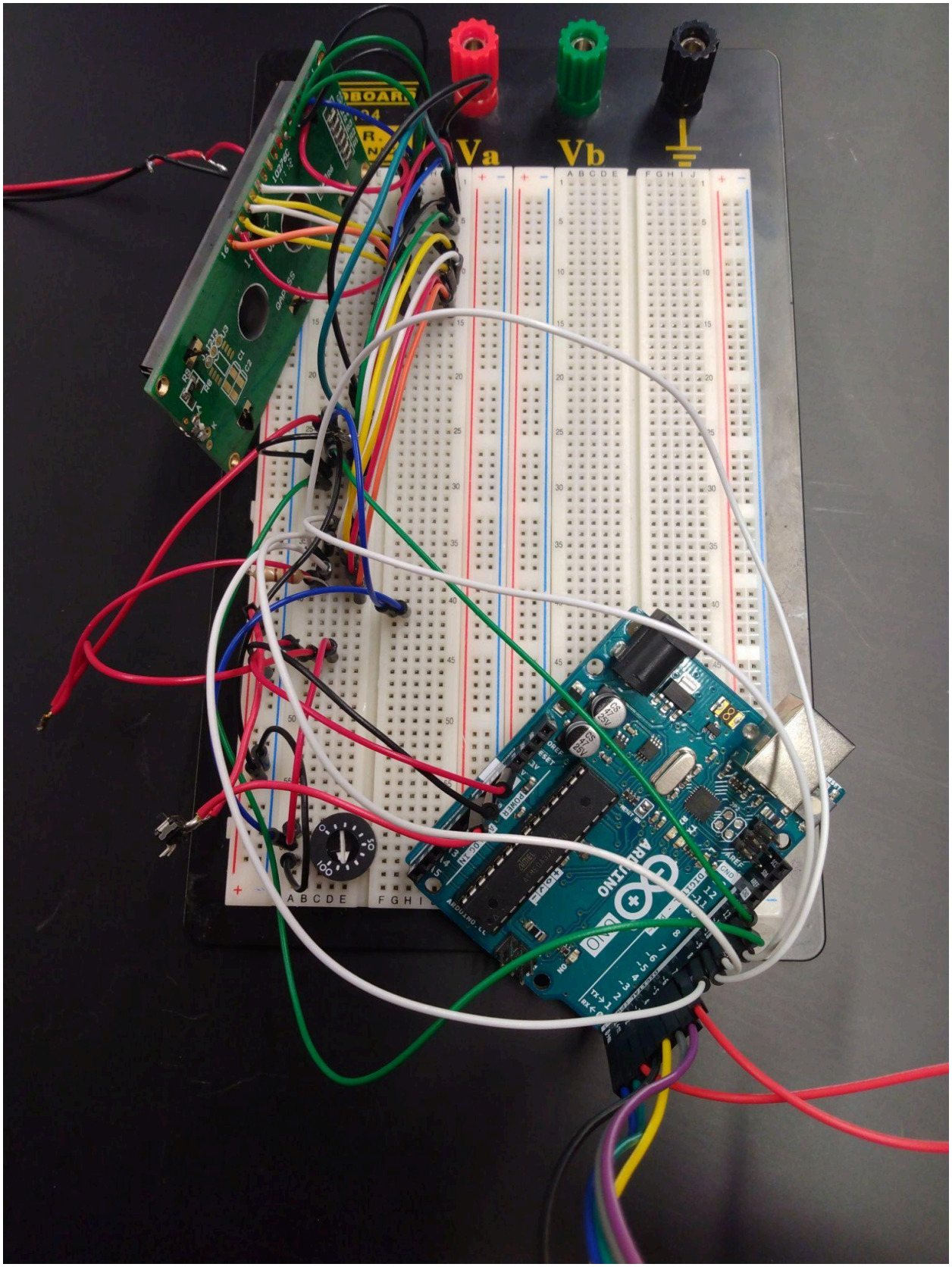
1. Connect to 5 volts, ground and contrast pin on LCD Display (Follow the diagram)

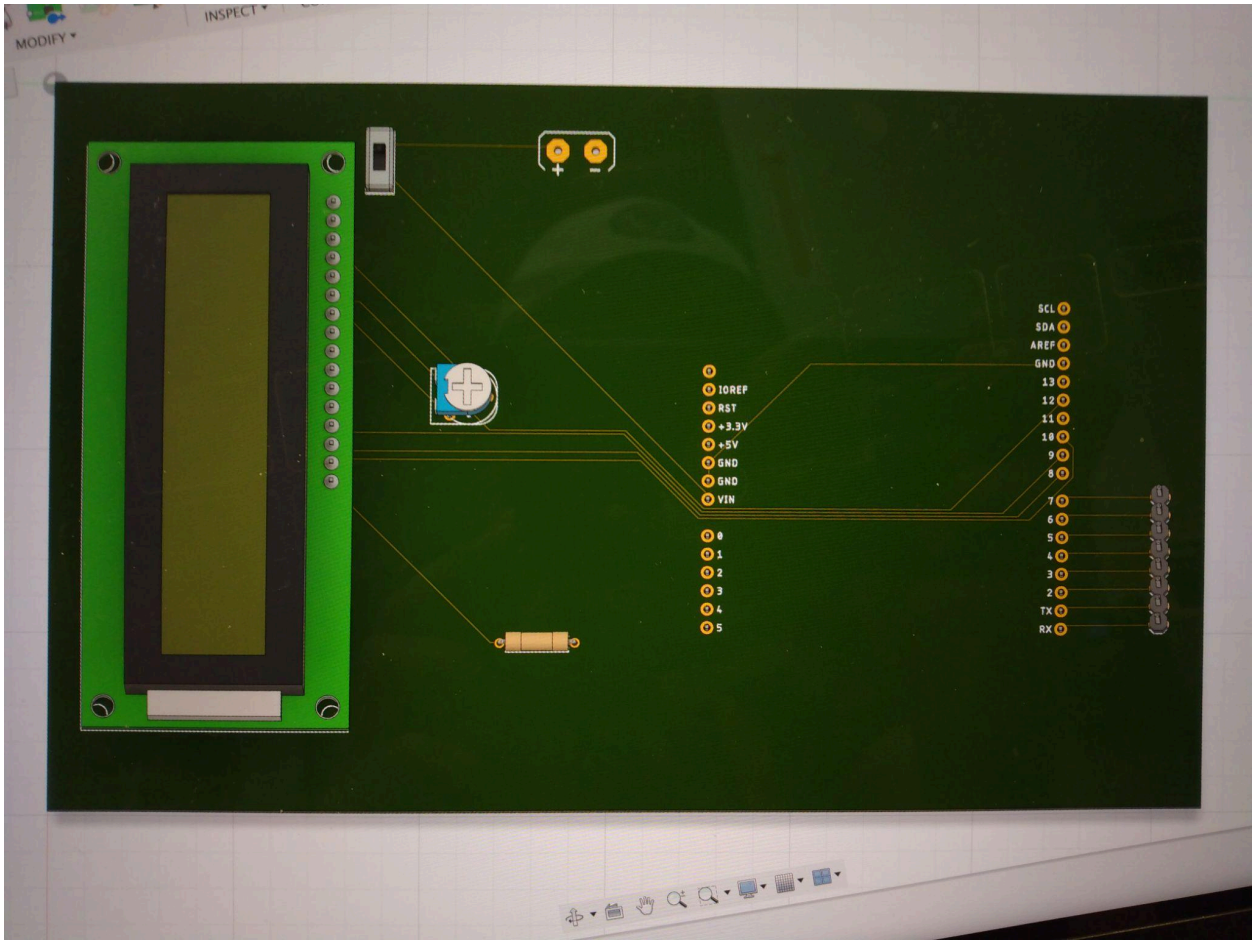
5. Power Switch
    a. Wiring
6. Battery
    a. Wiring
        1. Connect the 9 volt battery to the barrel plug battery clip.
        2. Connect the barrel plug to the arduino.
7. PCB Board
    a. Soldering
        1. Solder all the components to their respective places.
8. PLA Housing
    a. Assembly
        1. Assembly correctly please.

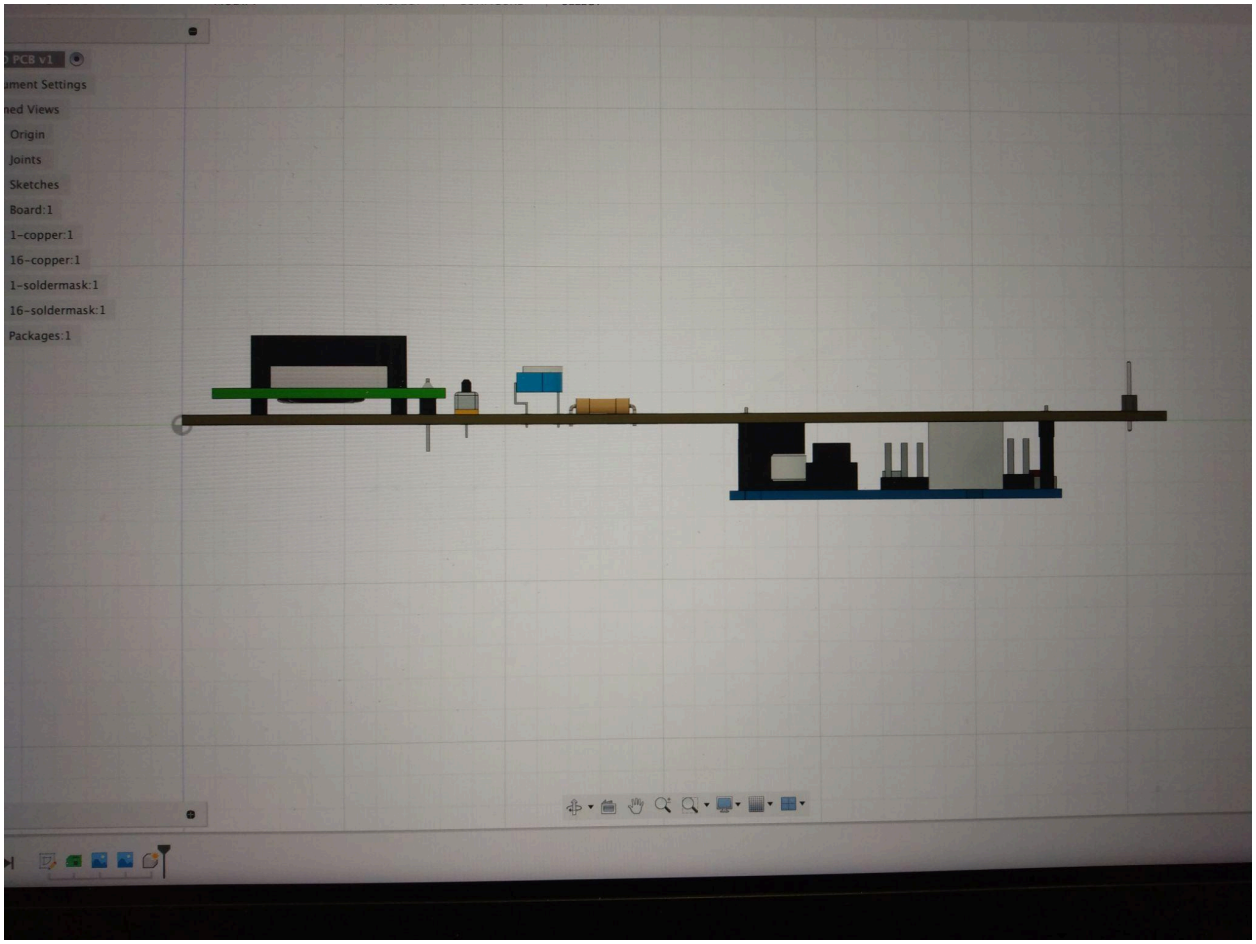
Data:

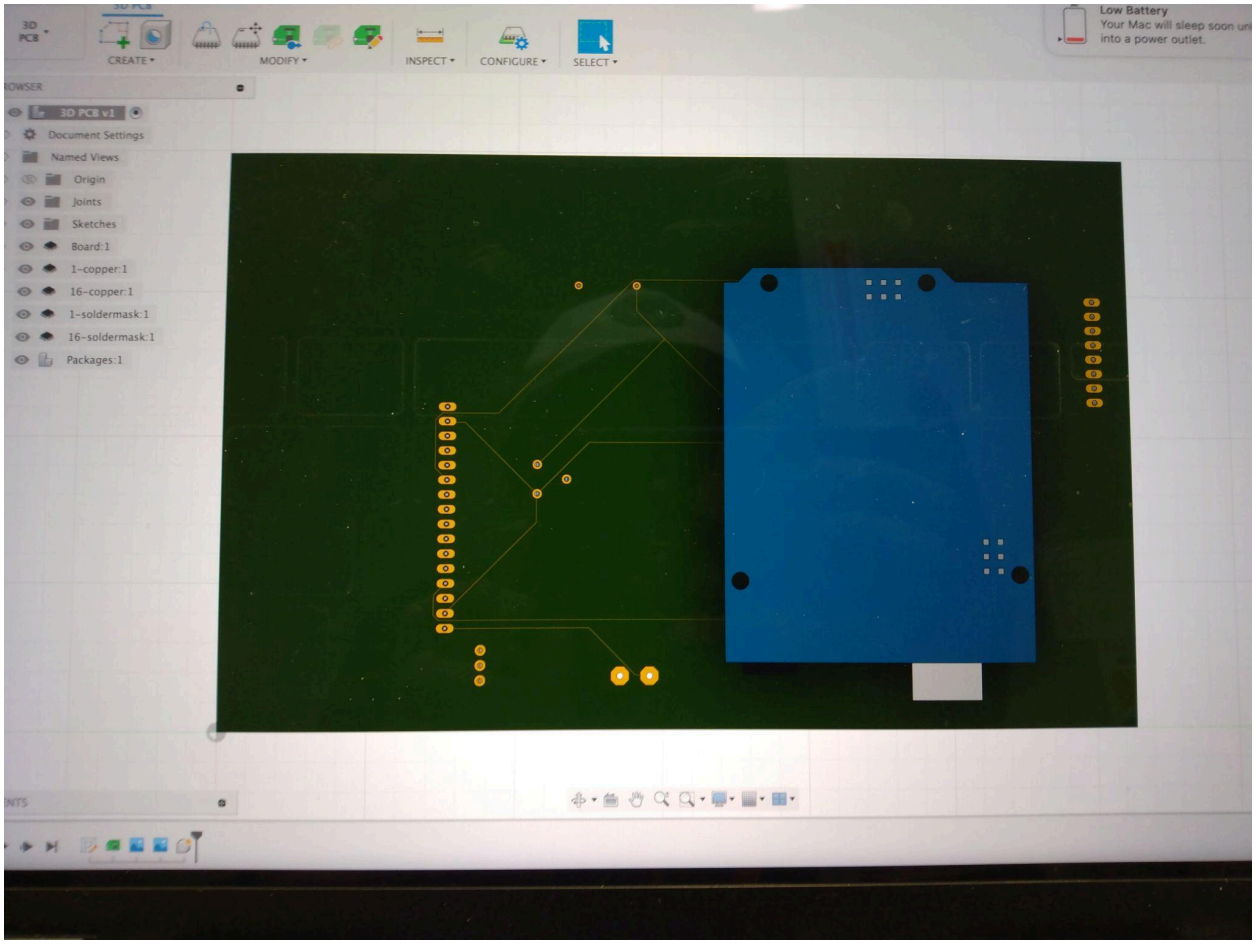



LCD1
LCD
Vss Vcc Vo RS R/W E DB0 DB1 DB2 DB3 DB4 DB5 DB6 DB7 LED+ LED-
R1
220Ω
Arduino
3V3 5V Power Vin
RST
AREF
IO REF
N/C
D13 D12 D11 D10 D9 D8 D7 D6 D5 D4 D3 D2 D1 D0
PWM
Digital Input/Output
A0 A1 A2 A3 A4 A5
Analog Input
GND
SCL
SDA
microcontroller-project.com
Connect Potentiometer
for fading. Adjust
Contrast.
7 8 9 /
4 5 6 x
1 2 3 -
0 = +


Va
Vb
ARDUINO


MODIFY
INSPECT
IOREF
RST
+3.3V
+5V
GND
GND
VIN
SCL
SDA
AREF
GND
13
12
11
10
9
8
7
6
5
4
3
2
TX
RX


PCB v1
ument Settings
ned Views
Origin
Joints
Sketches
Board:1
1-copper:1
16-copper:1
1-soldermask:1
16-soldermask:1
Packages:1


3D PCB
CREATE
MODIFY
INSPECT
CONFIGURE
SELECT
Low Battery
Your Mac will sleep soon un
into a power outlet.
3D PCB v1
Document Settings
Named Views
Origin
Joints
Sketches
Board:1
1-copper:1
16-copper:1
1-soldermask:1
16-soldermask:1
Packages:1




1 2 3
4 5 6
7 8 9
0

